

Indian requirements also plays a challenging role.

Lack of trust among customers has a huge impact on the business. The main reason behind this is uncertainty regarding quality of products. Dumbre informs, "There are hundreds of kits, but very rarely you see a kit that is well-thought-over and well-made. That being said, it is a huge opportunity for us. We will certainly see some big players in this segment in the coming two to three years. Indian companies are also trying to make their own kits, but none have been very successful so far."

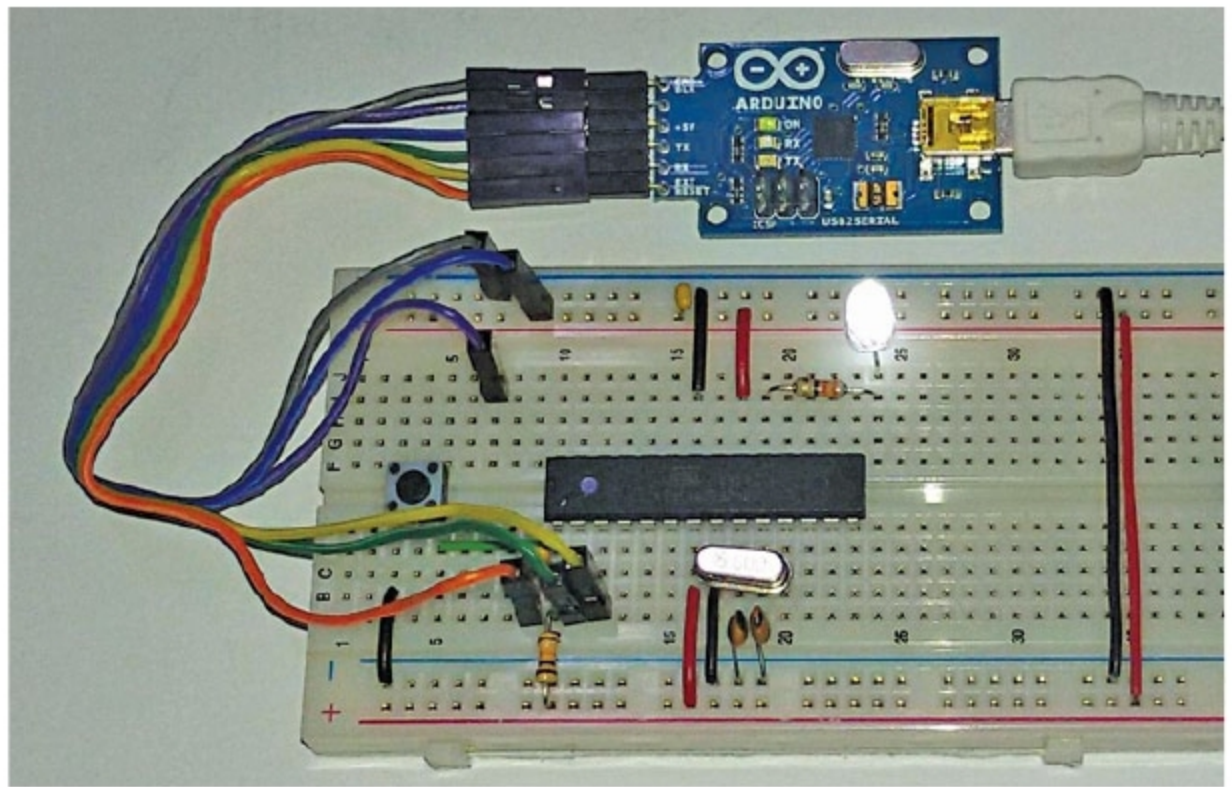
Due to these reasons, when it comes to globalisation of business, V. Ravi, chief operating officer, Anshuman Tech Pvt Ltd, says, "International marketing continues to be a challenge for Indian companies due to lack of financial incentives. You will see German, Korean, Chinese and UK pavilions at most expos, while Indian pavilions are conspicuous in their absence, and even if these exist, they can be seen only in tier II and III expos."

Another issue that needs to be discussed is availability of kits in different locations. A large population of India lives in rural areas. Experts agree that there is a huge gap in the urban and rural markets due to last-mile connectivity. Parab tells, "Manufacturers and the marketing sector in the field of educational kits are unable to reach district level in the country. Most parts of the country, especially rural regions, are not aware of the availability and, if that is not the case, they have to depend on metro cities to avail of these kits."

Barman says that there is a need to provide training in educational institutions with our own designed training products based on Indian school and college syllabi.

Demand and supply

Kits are becoming advanced nowadays. DIY kits like Arduino, due to their simplistic use, are very popular among beginners. The integrated development environment (IDE) for Arduino is suitable for high-level programming languages like C and C++. Plug-and-play modules, known as shields, can be bought for adding inbuilt functionality. Even Raspberry Pi, which was developed in the UK to teach children programming and physical computing,



A circuit built on a breadboard for an Arduino-based project

has a growing demand in the country.

About technology trends, Wagh says, "Demand for training products is increasing with the number of institutions and universities coming into existence. Supply side also has different variants and available features. With the need for AI, the IoT and computing requirements coming into major streams, there will be growth of educational kits for experiments required for educational institutions."

Although, availability of training products is sufficient as it is a mix of Chinese and Indian products, according to Barman, supply of high-quality products is not enough to meet the requirements when it comes to local dealers.

Parab adds, "Still, there is a huge difference between supply and demand not only in terms of quantity but also proper products to fulfil the exact requirements of the end customer, that is, students. Lack of a proper setup for after-sales backup is also a key issue in this field."

Jain agrees, "The latest kits are not available to kids. In India, we still get smaller kits that work on LM555 or LM324. The supply of Arduino or Raspberry Pi kits is less. But market is growing, and supply will increase in the coming years."

Changing scenario

The change in the ways of imparting knowledge to students over the past few years is apparent. Dr Dhekale

says, "Students get more knowledge in a short time with the use of LCD projectors and screens. Educational videos can be played on smart TVs using Wi-Fi facilities. Video cameras are used to record lectures, which can be broadcast with a TV relay station."

With growing awareness, demand for traditional products is also reducing. Manufacturers are upgrading their kits to compete in the market. Dumbre says from experience, "This field is very fast-changing. Trends come and go every year. The only exception is the projects in the education curriculum. For example, engineering students still use 8051 controller kit—and they have been using it from almost 15 years."

Market opportunities

Because there are many opportunities, new entrants just have to pick the one that matches their expertise. V. Ravi, chief operating officer, Anshuman Tech Pvt Ltd, suggests, "Upgrading traditional training products with the latest technology like Wi-Fi interface, PC interface, mobile interface, cloud-based delivery, etc should be on the menu to create differentiation in the already-crowded marketplace."

Sharing his opinion on starting a business, Gala says, "New companies should first do market research for the products in demand. Then, they should source raw materials in bulk and use advanced manufacturing technology to gain economies of scale. They should also focus on quality."